Time-to-Digital Converter Based on FPGA With Multiple Channel Capability

J. Torres, A. Aguilar, R. García-Olcina, P. A. Martínez, J. Martos, J. Soret, J. M. Benlloch, P. Conde, A. J. González, and F. Sánchez

***Abstract*—This contribution describes an accurate approach implementing a Time-to-Digital Converter using a Field-Programmable Gate Array (FPGA) device. Time differences with a FWHM better than 100 ps for 24 pairs of channels working simultaneously have been achieved. This was possible through the proper management of FPGA internal resources and by an accurate device calibration process minimizing the effect of temperature and voltage fluctuations. The system calibration results and the time differences between multiple channels are presented. The current approach suggests the possibility of carrying out precise Time of Flight (TOF) measurements with, for instance, Positron Emission Tomography (PET) systems.**

***Index Terms*—Delay lines, field programmable gate arrays, positron emission tomography, time measurement.**

I. Introduction

THE improvement that Time of Flight (TOF) measurements can provide for image quality in Positron Emission Tomography (PET) systems was experimentally demonstrated several years ago [1]. Due to the complex method suggesting implementation of electronics to achieve picosecond resolutions, there exist very few PET systems, which enable this TOF capability [2].

Reconstruction approaches in conventional PET systems only record that the two photons detected are within the same time coincidence window and, therefore, belong to the same positron annihilation event. Such an event is registered along the line connecting the two coincidence detected impacts with the same emission probability. However, in PET systems enabling TOF capabilities, it is possible to measure the time arrival difference of the two resulting gamma rays, providing a better localization of the annihilation event along the line connecting the two detectors. Nevertheless, the position is blurred by a time measurement uncertainty, termed time resolution.

There are several approaches to electronically implement TOF measurements. Most of them are based on Application Specific Integrated Circuits (ASICs), which offer a great advantage in terms of recurring costs as very little material is wasted due to the fixed number of basic elements in the design. Although the recurring cost of an ASIC is quite low, its non-recurring cost is relatively high. Another major drawback of ASICs is that they are nearly unreconfigurable, in addition to needing to be produced in high volumes to be cost-effective [3].

In recent years, the development of very high performance Field-Programmable Gate Arrays (FPGAs) has allowed for their successful application in a wide spectrum of areas [4]. FPGAs present high-speed data transfer, fast processing capabilities and the ability to simultaneously handle a huge number of electronic signals. Moreover, they enable the possibility of reconfiguration to adapt to a very wide range of applications without the need for modifying the hardware design. All these features have served to promote FPGAs as ideal candidates for the development of complex electronic systems.

FPGAs internal structures allow one to accurately determine the TOF of gamma rays, improving the signal to noise ratio of the reconstructed image [4]. The technical feasibility of using FPGAs to determine PET photons TOF with very accurate time resolution has been demonstrated [5], contributing to the improvement of commercially available PET systems (around 600 ps) [2]. The most significant advantages provided by FPGAs to PET systems are the high number of possible input channels and their huge data processing capability, allowing the implementation of complex detector coincidence maps in order to improve the true coincidence detection ratio. Additionally, the FPGA reconfiguration capability could allow for rearranging the detector setup without the need of a hardware redesign and in doing so becoming a cost effective solution.

Regarding TOF estimation, a possible approach to cope with time measurements below 1 nanosecond might be the use of high-speed clocks in the order of few GHz. Nevertheless, due to limitations in the bandwidth, this solution is not feasible in most currently available FPGAs [5].

The possibility of implementing Time-to-Digital Converters based on FPGAs has already been demonstrated [5]–[7]. Current FPGA TDCs relies on digital components such as buffers and flip-flops. There are several possible approaches, for instance, those that rely on the use of internal GHz ring oscillators, which are enabled when an input signal is received and the number of cycles counted. That is the case of [8], in which the purpose is to measure the width of a pulse and a resolution below 100 ps was achieved with the Altera Straix II device. Another common techniques take advantage of the internal logic gates, interconnecting several of them forming a chain, for which the

Manuscript received February 27, 2013; revised June 25, 2013 and September 13, 2013; accepted September 19, 2013. Date of publication November 11, 2013; date of current version February 06, 2014. This work was supported in part by the University of Valencia.

J. Torres, A. Aguilar, R. García-Olcina, P. A. Martínez, J. Martos, and J. Soret are with the Communications and Digital Systems Design Group (DSDC) Universidad de Valencia, Valencia 46010, Spain (e-mail: jose.torres@uv.es).

J. M. Benlloch, P. Conde, A. J. González, and F. Sánchez are with the Institute for Instrumentation in Molecular Imaging (I3M), Universidad Politécnica de Valencia-CSIC-CIEMAT, Valencia 46022, Spain.

Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TNS.2013.2283196

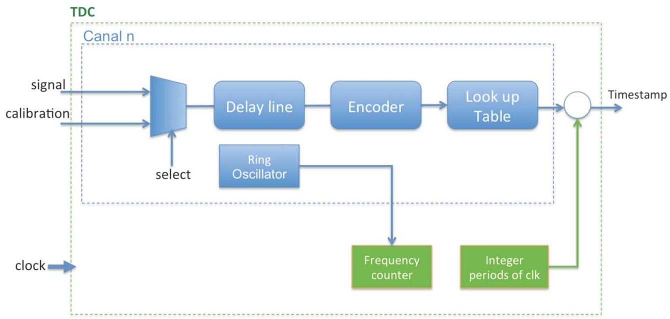

Fig. 1. Basic scheme of the implemented TDC.

individual delay is measured and the number of taps reached by the input signal counted.

In [5] a Virtex-5 FPGA is used to build a pipelined interpolating TDC. A counter is used for coarse measurements and a tapped delay line for the fine time ones. The achieved TDC resolution resulted on 17 ps. This solution is intended to be used as a high throughput time difference discriminator for optical communications. In [6] single tapped delay line interpolator in a Virtex-5 device is also used, obtaining a time resolution below 200 ps.

Another approaches, like [8], use multiple channels based on delay lines in combination with a technique called Wave Union, which minimize the ultra-wide bins problem by avoiding them. With such an alternative, a bin resolution of 10 ps has been achieved using an Altera Cyclone II FPGA device.

One of the drawbacks of a TDC-FPGA implementation is their sensitivity to voltage and temperature variations, which cause problems in system performance [5]. The problems relating to Process of Voltage and Temperature variations (PVTs) are also reported together with some strategies for assuring that their impact on the system performance is negligible.

This contribution presents the procedures carried out to accurately measure time differences with a resolution better than 100 ps and for multiple channels using a FPGA device. The suitability of a fine time measurement procedure for TOF calculation in PET systems will also be shown. Following an overview of the system architecture, the characterization of the TDC is described, focusing on aspects such as its calibration and PVT compensation. Then the implementation details of the experimental set-up used for time resolution measurements as well as its analysis will be described, followed by the conclusions and final remarks.

II. TDC Architecture

The system architecture presented in this work relies on a Time-to-Digital Converter (TDC) core generating timestamps for every incoming pulse signal. The TDC architecture scheme is presented in Fig. 1.

The goal of the TDC is to recognize events and provide a digital representation at the time they occurred. The current implementation of the TDC core is based on a modified Vernier delay line, better known as Nutt structure [9]. The delay line is in charge of getting times under the clock period. Fig. 2 shows a basic delay line structure with logic gates connected to form a chain.

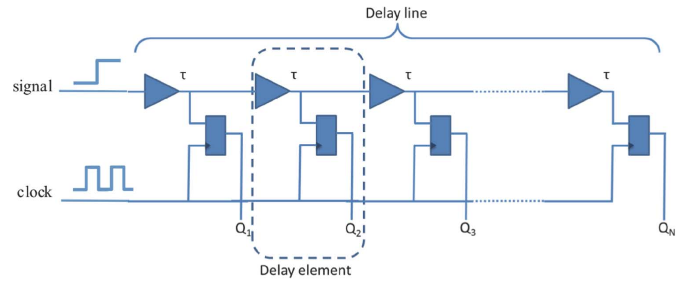

Fig. 2. Delay line scheme.

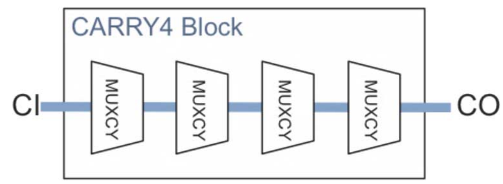

Fig. 3. Sketch of the Xilinx CARRY4 block.

Each delay element (also known as tap) corresponds to the MUXCY element of a CARRY4 block, found in several Xilinx FPGAs (see Fig. 3) [10]. The MUXCY can be modeled as a logic gate with certain time propagation so that the input signal is delayed as long as the total time of all taps belonging to the delay line. The goal of using a carry chain is that the blocks can only be placed along the longest physical dimension of the device. Once the first element is placed, the location of other taps is almost automatically determined by the tools used to program the device. Because of this, the delay line design is easily portable to other devices as long as they have this type of structure.

The lower the delay of each MUXCY element, the better time resolution achievable. Depending on the device selected and its speed grade, these times vary. Normally, propagation times are not presented as MUXCY individual delays in Xilinx datasheets but for the complete CARRY4 block delay. The time is designated by T_{BYP} and it is described as *CI input to CO output*. For instance, a Xilinx Spartan-6 device has a maximum T_{BYP} of 100 ps for a -2 speed grade type. Since each CARRY4 block has 4 MUXCY elements, the average delay for one MUXCY element is about 25 ps. In the case of a Kintex-7 Xilinx device, the maximum T_{BYP} is 60 ps, also for -2 speed grade. Consequently, the delay average of each MUXCY is about 15 ps, significantly lower than the Spartan device [10].

Once the signal crosses the delay line, the number of taps reached is counted (what it is also known as thermometer code) by an encoder and converted to time by means of a Look Up Table (LUT), which has previously stored the equivalencies in a calibration process (the details are presented in Section III). The timestamp is finally generated using a combination of the integer number of clock periods and the fine time provided by the LUT.

III. TDC Characterization

In this work a 24 channel TDC has been implemented on a Kintex-7 device, more specifically on the xc7k325t-2ffg900 model. A 200 MHz oscillator with a frequency jitter of 20 ppm has been used as the system clock. The 24 channels distributed

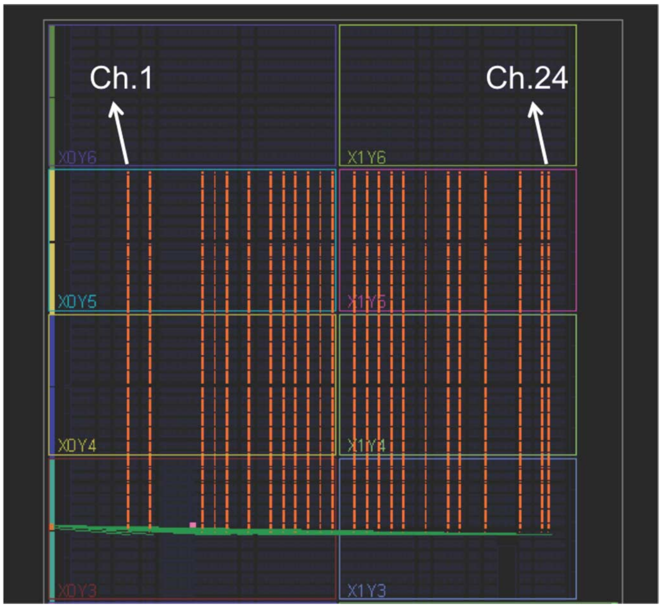

Fig. 4. Distribution of channels inside the used Kintex-7.

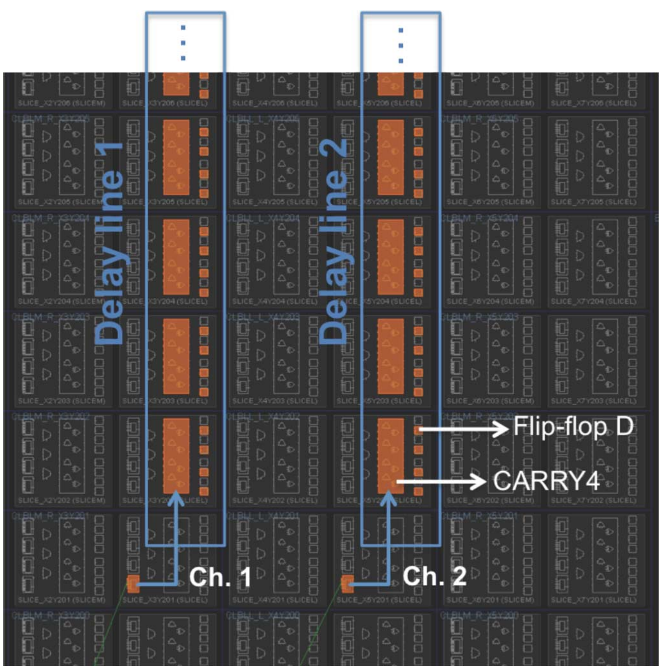

Fig. 5. Detail structure of two channels. When the system clock enables the output of D flip-flops, the position in the delay line of the incoming pulse is recorded.

inside the FPGA are depicted in Fig. 4. Here, the dashed orange blocks are the CARRY and sequential logic, while green wires at the bottom are the connections between the input signal and the delay lines.

It is convenient to note that only the delay lines are shown in Fig. 4 in order to clearly visualize the 24 channels and, for that reason, other resources are hidden.

Fig. 5 shows the details of two channels. It exhibits the scheme of the modified Vernier delay line already presented in Fig. 2, in which each tap has its own flip-flop D. Although it is not shown in this figure, the output of each carry block is connected with the next carry immediately above.

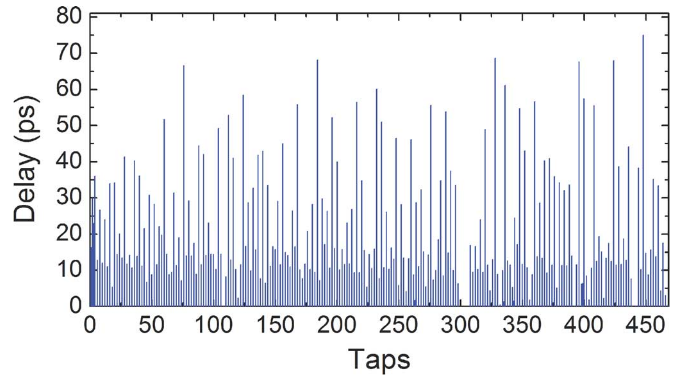

Fig. 6. Delay line histogram of the first channel.

A. Initial Calibration

An initial calibration is required since the delay of each tap must be determined in order to characterize the whole delay line of each channel. A ring oscillator based on 41 NOT gates has been implemented, resulting a ring frequency of around 30 MHz. Apparently, it should be enough to use only one ring oscillator for all channels but, due to the necessity of compensating for the PVT processes, an individual ring oscillator per channel is devised to compensate more accurately for the variations of each delay line (see the "C" section for more details).

The ring oscillator output is initially injected into the delay line with the purpose of measuring the events registered by the taps: each ring oscillator pulse is propagated through its corresponding delay line and only its rising edge is taken into account. When the system clock rising edge enables the output of D flip-flops, the last tap reached by the input signal is stored. Hence, the longer the delay of the tap, the more events it will register. If the sum of the delay taps is higher than the system clock period, there will be a tap from which no events will be registered because the system clock period has completed it already. Then, by sending pulses asynchronously, registering the taps in which the input event is and knowing that the last tap reached corresponds to a delay equal to the system clock periods, it is possible to compose a histogram of events and extrapolate the delay of each tap. This process is known as a code density test [11]. Due to the asynchronisms between the ring oscillator and the system clock, the entire delay line can be covered. Once each delay is determined, the values are stored in a LUT, enabling the TDC to start the measurements. The time that it takes to calibrate the delay line depends on the number of pulses that are defined. 200 000 pulses have been used, which results in a time of roughly 29 ms per channel since the logic resources to determine the values of the LUT are shared. It should be remarked that this calibration process is only carried out once, before the TDC starts to generate timestamps.

After the initial calibration process, the individual delay time of each element belonging to each delay line is determined, as depicted in Fig. 6, and time values are stored in the internal device memory.

The last tap reached for all the incoming calibration pulses shows that the total amount of time corresponds to the system clock period. For the system clock period of 5 ns, the last tap reached was 463, which means that the accumulated delay corresponds, precisely, to the system clock period. As the system

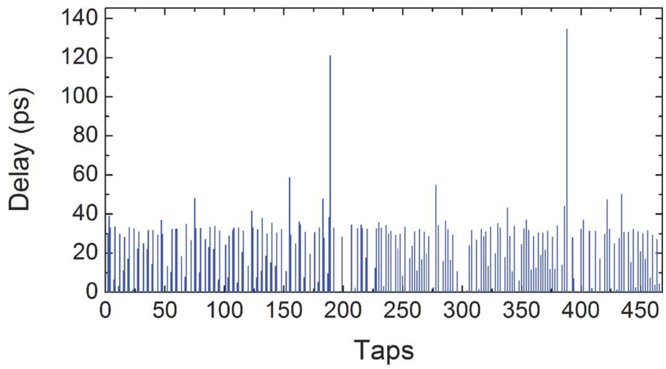

Fig. 7. Delay line histogram of the channel shown in Fig. 6 after the readjustment process.

clock period decreases, the number of required taps decreases as well. The more taps, the more space used on the FPGA, the more complex the design becomes and the higher accumulated jitter results [11]. The use of a higher system clock frequency was not permissible due to the constraints of the system some of which could then be compromised. Then, according to the Kintex FPGA model used, it was determined that the delay line length does not compromise our requirements in terms of number of channels and, thus, the decision was made to adopt the model presented. It is important to highlight the fact that almost the half of the taps do not present any delay at all. This might occur because of the intrinsic structure of the CARRY4 blocks and remain an unreconcilable hindrance.

Fig. 6, readily shows that time delays are not homogeneous. Depending on the period in which the incoming signal falls, the time accuracy can vary significantly. Therefore, different time events may not be registered with the same accuracy. The origin of delay discrepancies between taps seems to come from the CARRY4 structure. It was observed that, according to the carry chain distribution, some taps, which are placed far from others in a single CARRY4 block, register the event coming from the input signal first. This is due to the "look ahead" feature of the CARRY4 blocks, which is supposed to improve the CARRY logic and which allows some taps to accomplish their task easily. That might be the reason for the appearance of the bubbles mentioned in [5]. Another explanation can be found in the clock path differences that trigger the D flip-flops of the delay line. If those differences are notable, asynchronisms between them will appear and may cause the effect mentioned. In order to avoid this, the thermometer output was analyzed as well as the value of the flip-flops. The taps that are located nearest in the delay line and have registered the event later than the ones located further are realigned, giving more uniformity in the encoder counting process. Once these corrections were addressed, the initial calibration process was launched again, obtaining the histogram showed in Fig. 7.

An improvement in the homogeneity of the tap delays is observed. Except for some taps (those with longer delays due to the longer path until the next tap) the histogram presents a narrower distribution value. Specifically, the Fig. 6 presented a standard deviation of 17.8 whereas in the Fig. 7 the value decreases until 12.3 ps. As has been mentioned, this will allow the improvement on the achievement of specific time resolutions independent of the delay line period in which the input signal falls. In the following subsection, the homogeneity details resembling those from this last histogram are studied.

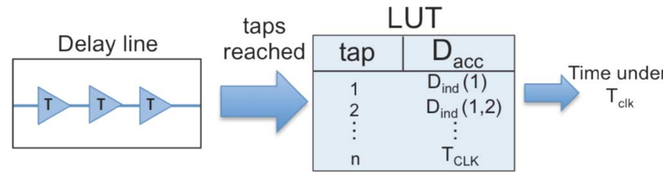

Fig. 8. Generation of times under T_{CLK}.

The individual delay D_{ind} of each tap can be determined using the following equation:

$$D_{\mathrm{ind}}(n) = \frac{H(n+1)}{C} \cdot T_{CLK}, \tag{1}$$

where H is the number of hits registered in the n tap, C is the total amount of generated calibration pulses and T_{CLK} the system clock period. Once the individual delay is known, the time accumulated until the tap n has to be determined from the (2).

$$D_{acc}(n) = \sum_{i=1}^{n-1} D_{\mathrm{ind}}(i). \tag{2}$$

Finally, the calibration process ends with the storage of the time values in a LUT, in which the fine portion (times below system clock period) of the timestamp is composed. Fig. 8 illustrates this process.

B. DNL and INL

The DNL is the difference of the bin size from its ideal value normalized to the ideal bin size value. The integral nonlinearity (INL) corresponds to the cumulative addition of the DNL values. By a proper tap selection, it is possible to reduce the DNL [6] but then the design becomes placement and routing dependent and this difficult the portability to other devices. Consequently, the decision was made to maintain the carry chain structure, taking into account about half of the bins since the others had such a short delay that was negligible in comparison.

In Fig. 9, graphs for both cases, both DNL and INL, are depicted. These values have been taken from the channel that presented the worst-case values. For a system clock period of 5 ns, the bin size is supposed to be 22.7 ps. For the differential non-linearity a value of less than 3 Least Significant Bits (LSBs) and an $\mathrm{INL} < 4$ LSB, both peak to peak, has been achieved. Taking into account all channels, the mean value of DNL is 2.6 LSB and 3.4 LSB for INL. Moreover, the standard deviation obtained for the 24 channels has been of 0.8 LSB in the case of DNL and 0.6 LSB for INL.

C. PVT Compensation

When the delay line is characterized, a second calibration process takes place at the same time that the system provides timestamps. This is necessary since PVTs may affect the delay features of the carry chain [12].

In some contributions, additional delay lines strategically placed apart from the ones in charge of timestamp determina-

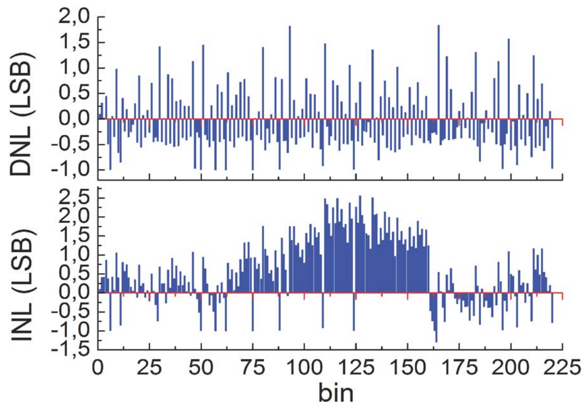

Fig. 9. Top, DNL measurement plot. Bottom, INL measurement plot. Both for the worst channel case.

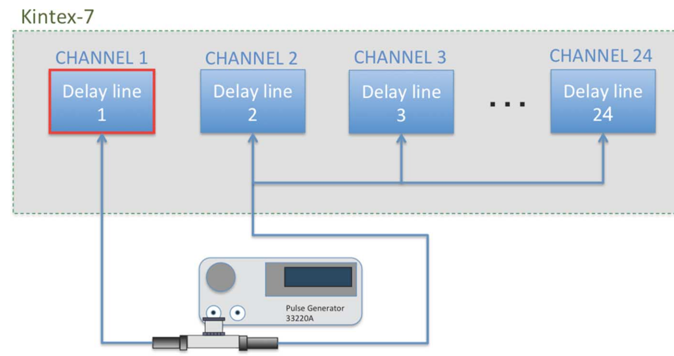

Fig. 10. Configuration scheme for the 24 channels.

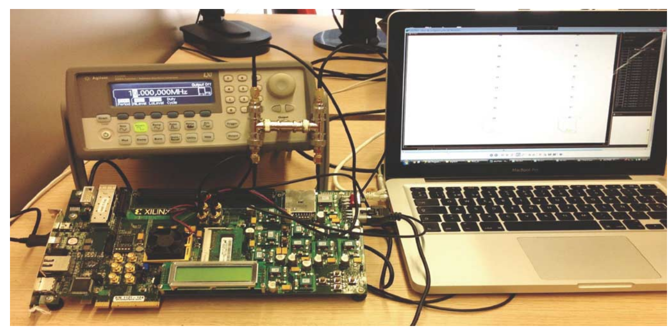

Fig. 11. Measurements set-up. KC705 Kintex-7 board, function generator and data acquisition.

tion are implemented for PVT compensation using the stable reference of the system clock period [13]. With this principle in mind, corrections to the timestamps can be made, for instance, by an adjusting the device voltage supply or through compensation of the stored bin time values.

In this work, the same ring oscillator used for initial calibration is reused for this purpose. While the system is working, the ring oscillator frequency is measured and compared with that which was stored at the beginning of the measurements. As the tap delay varies with the fluctuations of voltage and temperature [6], the ring oscillator frequency will vary with PVT, thus making it possible to extrapolate its effect on the delay line interpolating the differences of frequency values of the delay elements previously stored and overwriting them with new ones. In the Results section, Subsection B, the details of the experiment carried out are presented.

D. Dynamic Range

The TDC core generates timestamps, the widths of which are 32 bits: 20 for the system clock periods and 12 for the fractional portion, which means that for a system clock period of 5 ns, the LSB corresponds to 1.22 ps and the measurement range to up to 5.24 μs. This is large enough for the application, since it does not require measurement of times above few hundred nanoseconds. In fact, a 32 bits timestamp is adopted because of the microprocessor register length that is used, which is precisely this value.

E. Deadtime

Between the arrival of the event to the TDC and generation of a timestamp, a certain amount of processes take place which induce deadtime. Among them, the propagation time until the event is registered (less than one clock cycle), the encoding process, the LUT access and the final operations to shape the timestamp. The total time is 30 ns for a system clock period of 5 ns, which determines the system deadtime. This time is not critical for the main application of the designed TDC, in which data with a periodicity above 100 ns for the incoming signals is expected.

IV. Implementation

In the following set-up, the usage of the new TDC for a possible PET system with up to 24 detectors with TOF calculation capabilities will be described (see Fig. 10). The signal coming from the pulse generator was injected into a taped delay line implemented in the Kintex-7. The output was divided into two identical signals in order to simulate the arrival of true coincidences coming from different detectors forming a PET system. Internally, in the FPGA, one of these signals is split into 23 more signals and each of them connected to one of the remaining delay lines.

The various delays are measured with respect to channel 1. Since the channels are placed along the FPGA (see Fig. 4), the higher the channel number the longer internal routing path associated and, thus, the longer time difference.

One of the main advantages of analyzing these time differences is that the input pulse is the same for all channels suppressing signal jitter and making the results independent of this effect. Doing the experiment in this way makes it possible to analyze the TDC core independently of the input signal jitter, allowing a precise characterization.

Fig. 11 illustrates the set-up used during the measurements. The signals coming from the pulse generator were injected into the Xilinx KC705 board through SMA connectors. A 12 MHz, 1 V_{pp}, $V_{off} = 0.5$ V_{dc} square signal was generated. TDC timestamps were serial with the JTAG port to a computer, which receives and presents the data using the software ChipScope, which is an internal logic analyzer developed by Xilinx. This was necessary in order to analyze the more than 400 bits generated.

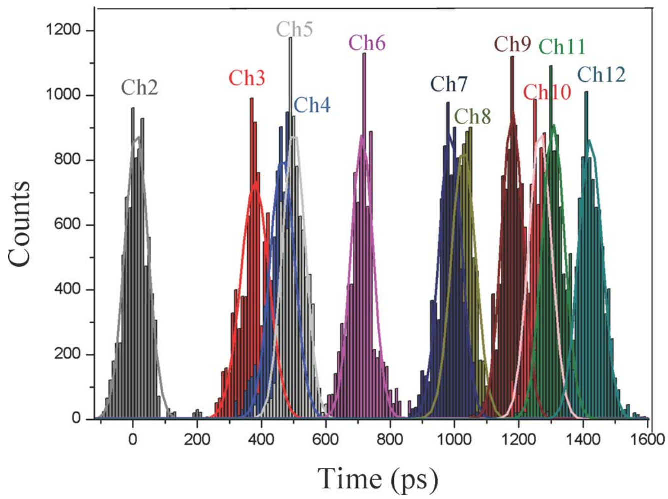

Fig. 12. Measured time differences between channel 1 and channels from 2 to 12.

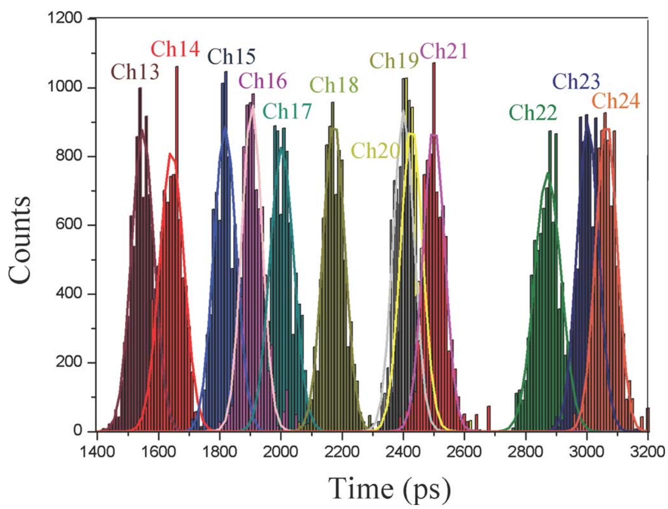

Fig. 13. Measured time differences between channel 1 and channels from 13 to 24.

V. Results

A. Time Differences

Figs. 12 and 13 present the time differences between the first channel and the other 23. The profiles have been fitted to a Gaussian distribution. It is expected that the mean of each histogram would rise with the increasing number of channels. These results are generated with a total amount of around 6 500 samples each.

In Fig. 12 the location inside the FPGA of each channel (2 to 12) can be seen through the spaces between each histogram. The first histogram (time difference between channel 2 and channel 1) has its centroid at 10 ps while the last is centered at 1422 ps. Fig. 13 shows the results for channels 13 to 24. In this case, channel 13 has a time delay with respect to channel 1 of 1547 ps while channel 24 shows a difference of 3063 ps.

Fig. 14 represents the FWHM of all channels and their differences respect to the average value. Analyzing the FWHM values of all channels, it can be observed that, regardless of the point of the delay line on which the input signal falls, the FWHM values are very similar. In particular, the minimum value is 78.2 ps whereas the maximum only extends to 102.7 ps, making the total deviation of all channels as small as 6.5 ps. The average value of the FWHM is 85.7 ± 1.2 ps showing that the system performance is dependent neither on the introduced delay nor on channel position.

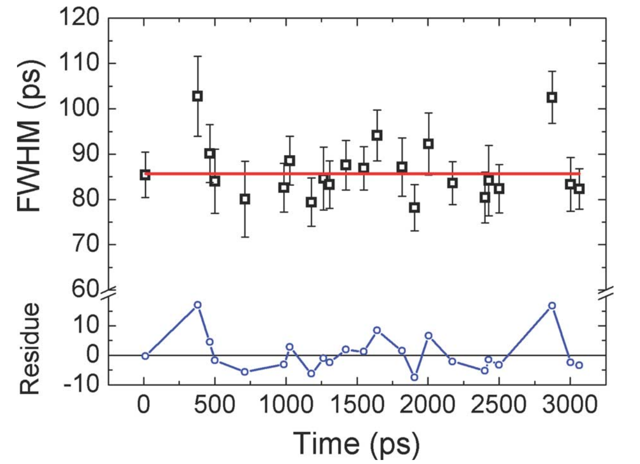

Fig. 14. Values of FWHM resulting of the time differences between the 24 channels.

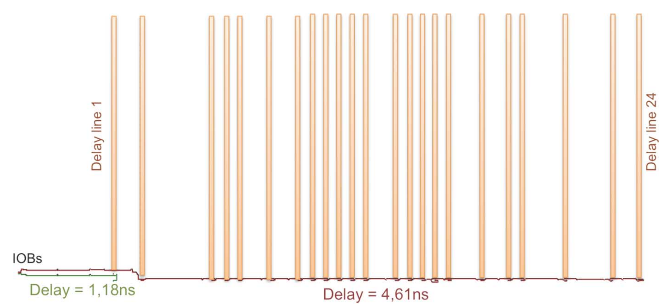

Fig. 15. Internal routing of the input signal to the delay lines. Times are measured with FPGA Editor tool.

In order to better understand the delay between each channel histogram, Fig. 15 plots an internal view of the Kintex-7 FPGA key delays measured with the specific tool FPGA Editor for Xilinx FPGAs. It also depicts the total delay of the signals coming from the Input/Output Block (IOB) to the delay lines (red bottom wire) as well as the delay from the IOB to the first channel (green bottom left wire), being 4.61 ns and 1.18 ns, respectively. The difference value (3.46 ns) is roughly the center of the histogram belonging to the channel 24 of Fig. 11. The remaining time differences can be extrapolated from the distance between the delay lines of Fig. 12.

B. PVT Compensation

In order to validate the compensation effects, the temperature of the internal FPGA die was measured through the internal on-chip sensor, which provides a digital value of it. The FPGA temperature was increased from 35° C to nearly 75° C by means of a climate chamber (Mod. ACS CH600C) with a ramp of 2° C/min. At the same time, pulses coming from the pulse generator used in the experiment of the last section were introduced into two TDC channels, numbers 1 and 2 (see Fig. 15), and the time differences between them calculated. Fig. 16 presents the results obtained. Time differences are Gaussian adjusted and their mean value evaluated. In order to test the compensation

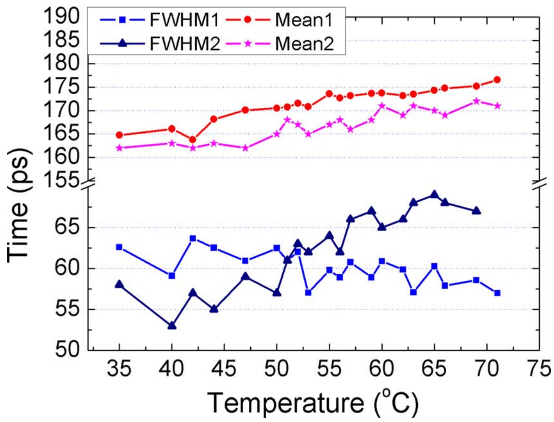

Fig. 16. Mean and FWHM values of the time differences corresponding to measurements with PVT compensation and without it.

effects of the suggested method, the time differences as well as the centroid values are calculated for two assumptions: one with PVT compensation and a second one without it.

The two lines on the top of the graph represent the Gaussian mean values of the time differences for both cases. The one with circles belongs to the case of PVT compensation whereas the one with stars belongs to the test without PVT compensation. Moreover, The two lines on the bottom of the graph show the FWHM of the time differences. The one with triangles and the other one with squares belong to non-compensation and compensation respectively. Fig. 16 shows the consequences of temperature variations on system performance for the two mentioned cases. For the one with PVT compensation, a low variation of the mean value with the temperature, in the order of 0.25 ps/° C (a 1.1% of the LSB), can be observed. However, this is not critical since the FPGA is intended to work under a temperature-controlled environment, in which the temperature will not fluctuate more than ±5° C. Regarding the FWHM of the time differences, the fluctuation in the measured temperature range is also negligible, these being less than 0.13 ps/° C. In contrast, in the case of turning the compensation off, the value of the time differences increases at the rate of 0.5 ps/° C. Therefore, for a ten degrees variation, it would correspond to 5 ps of increment, what it is considered interesting to compensate. The mean values shows roughly the same tendency than the first case and thus, they can be neglected.

As it was expected, the FWHM of the time differences increases with the temperature. The reason is the increment of the bin delay with the temperature, which makes the measurements more imprecise.

Moreover, although the voltage variations have not been characterized, their effect on the system performance is also compensated in the same way in as much as, the higher voltage supply the longer tap time delay [6].

VI. Conclusion

In this work, the FPGA capabilities for precise time difference measurements have been tested. It has been shown the possibility of measuring time intervals with better than 100 ps accuracy. The histograms presented in Figs. 11 and 12 resulted in an average FWHM of (85.7 ± 1.2) ps and 6.5 ps as the total FWHM deviation for the 24 channels. The results show the capability of multiple channels working simultaneously.

The initial calibration has been carried out to characterize the delay line, i.e., to determine the delay of each tap. Pulses coming from an internal ring oscillator have been used for this purpose and the same ring oscillator has been reused for PVT compensation, resulting in a low dependence of system performance on temperature and voltage. A DNL of 2.6 LSB and an INL of 3.4 LSB for a $\mathrm{LSB} = 22.7$ ps has been achieved for the 24 channels in an average, being the standard deviation of 0.8 and 0.6 LSB respectively. The DNL value can be improved through proper clock distribution inside the FPGA. The FPGA is divided into different clock sections, with an optimized clock path in order to minimize skew. Since the clock signal is in charge of triggering the delay line flip-flops, it may cause asynchronisms between them. Additional work is therefore required in order to evaluate such effects.

Concerning the deadtime and the delay line size, increasing the system clock frequency could reduce both. So, should the application need more data rate, its value could be modified. Additionally by reducing the delay line size, logic resources are gained as well as a reduction in the jitter introduced by the taps. Wave union techniques are usually implemented when the time between the delay line taps is significantly long [8], i.e., several times bigger than the average bin width. Concerning this design, wave union techniques would not introduce substantial improvement due to the adopted structure and the proper location of the carry chains.

Initial tests suggested the possibility of including more channels in a Kintex-7 type. In particular, up to about 64 channels might be allocated, since only 21% of device Slices have been occupied for the present test with 24 channels. The implemented TDC is intended to be used in a complete PET-TOF system [14], [15]. A time calibration process for such a PET system using photomultipliers tubes as photosensor and constant fraction discriminators for precise time information is currently being worked on.

Finally, the results here presented demonstrate the viability of using FPGAs for fine time measurements as TOF in complex PET systems, highlighting parameters such as the reconfigurability for implementing more channels and functionalities as well as the cost, this last being lower than a hundred of dollars for some models.

References

[1] S. Surti *et al.*, “Performance of philips gemini TF PET/CT scanner with special consideration for its time-of-flight imagin capabilities,” *J. Nucl. Med.*, vol. 48, no. 3, pp. 471–480, Mar. 2007.

[2] [Online]. Available: http://www.healthcare.philips.com/-Gemini TF PET/CT

[3] The Field-Programmable Gate Array (FPGA): Expanding Its Boundaries, InStat Market Research, Apr. 2006.

[4] M. D. Haselman *et al.*, “Digital pulse timing in FPGAs for positron emission tomography,” *CiteSeerX-Scientific Literature*, 2010.

[5] C. Favi and E. Charbon, “A 17 ps time-to-digital converter implemented in 65 nm FPGA technology,” *ACM 2009* [Online]. Available: http://infoscience.epfl.ch/record/139431

[6] C. Hervé *et al.*, “High resoulution time-to-digital converter (TDC) implemented in field programmable gate array with compensated process voltage and temperature (PVT) variations,” *Nucl. Inst. Meth. Phys. Res. , A*, vol. 682, pp. 16–25, 2012.

[7] S. Junnarkar *et al.*, “FPGA-based self-calibrating time-to-digital converter for time-of-flight experiments,” *IEEE Trans. Nucl. Sci.*, vol. 56, no. 4, pp. 2374–2379, Aug. 2009.

[8] Wu *et al.*, “The 10-ps wave union TDC: Improving FPGA TDC resolution beyond its cell delay,” in *Proc. IEEE NSS Conf. Rec.*, 2008, pp. 3440–3446.

[9] J. Kalisz, M. Pawlowski, and R. Pelka, “Error analysis and design of the nutt time-interval digitiser with picosecond resolution,” *J. Phys. E: Sci. Instrum.*, vol. 20, pp. 1330–1341, 1987.

[10] [Online]. Available: http://www.xilinx.com/support/documentation/data_sheets/ds182_Kintex_7_Data_Sheet.pdf

[11] H. Menninga *et al.*, “A multi-channel, l0 ps resolution, FPGA-based TDC with 300 MS/s throughput for open-source PET applications,” in *Proc. IEEE NSS Conf. Rec.*, 2011.

[12] J. Song *et al.*, “A high-resolution time-to-digital converter implemented in field-programmable-gate-arrays,” *IEEE Trans. Nucl. Sci.*, vol. 53, no. 1, pp. 236–241, Feb. 2006.

[13] J. Wu *et al.*, “Firmware-only implementation of TDC in FPGA,” in *Proc. IEEE NSS Conf. Rec.*, 2003, pp. 177–181.

[14] J. Torres *et al.*, “High resolution time of flight determination based on reconfigurable logic devices for future PET/MR systems,” *Nucl. Instrum. Meth. Phys. Res. A*, vol. 702, pp. 73–76, 2013.

[15] W. Gao *et al.*, Integrated High-Resolution Multi-Channel Time-to-Digital Converters (TDCs) for PET Imaging N. Laskovski, Ed., ISBN: 978-953-307-475-7.